

PROBLEM SET 3

AI-Native Field Address Geocoder

Exploratory Data Analysis, Model Architecture,
Evaluation and Delivery Report

Course:	Applied Data Intelligence (ADI) — PS3
Project:	Address-intelligence geocoder with uncertainty and evidence trails
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1. Executive Summary

CreditNirvana’s field-collections teams must find borrowers whose addresses are often written as descriptions rather than postal locations. A commercial geocoder may return a locality centre even when the useful operational answer is “behind the ration shop, two lanes after the temple.” The resulting search time is expensive, productive visits are lost, and an account can be marked address-not-traceable even though the borrower is present.

This project implements an address-intelligence service for that problem. It combines address text, landmarks, localities, historical visit evidence, GPS quality, dwell time, visit outcomes, remarks, integrity signals and shared-place evidence. The model is a 150-tree LightGBM classifier over a 21-feature evidence contract, fronted by a deterministic reliability-weighted aggregator that works even where the learned model has no support. The service returns a predicted location, an uncertainty radius, an explainable evidence trail and an operational recommendation. Reliable successful visits improve future predictions, while confirmed and predicted locations remain separate so that a weak prediction cannot overwrite a strong observation.

On the supplied train split (2,175 addresses, 3,896 visits, 66 surveyed ground-truth addresses), the evidence-weighted geocoder achieves a median location error of **204.4 m** compared with **380.8 m** for the commercial baseline geocoder — a 46.3% reduction in median error — and places 53.0% of addresses within 250 m versus 28.8% for the baseline. The uncertainty radius covers the true location for 72.7% of predictions, the expected calibration error is 0.155, and each prediction takes about 3.3 ms. The service is deliberately conservative: it uses projected PS3 coordinates as metres during evaluation, does not treat every check-in as ground truth, and refuses to report causal treatment effects because the supplied data lacks treatment assignment and an independent recovery outcome. Every reproducible evaluation run is recorded with a run ID, artifact version, configuration, source-file SHA-256 hashes, metrics and explicit limitations.

2. The Business Problem

2.1 Operational setting

The platform serves secured and unsecured retail, MSME and microfinance portfolios. A lender may hold millions of accounts across delinquency buckets, and field work is far more expensive than digital contact. The fully-loaded cost of the available channels is summarised in Table 1.

Table 1. Typical fully-loaded cost per contact channel.

Channel	Typical fully-loaded cost
SMS / WhatsApp	Fraction of a rupee
AI voice	Approximately INR 1–3 per call
Human tele-caller	Approximately INR 15–30 per connect
Productive field visit	Approximately INR 150–400

The correct objective is therefore recovery *net of contact cost, customer goodwill and compliance risk*. A geocoder is useful only if it increases the probability that a field visit is productive, without encouraging unlawful or excessive contact.

2.2 Why ordinary geocoding fails

Indian collection addresses frequently contain:

- mixed Devanagari, Roman, Kannada, Hindi and English text;
- transliteration differences such as *mandir* and *temple* for the same landmark;
- abbreviations, spelling errors and inconsistent house numbers;
- relational descriptions such as “behind”, “near”, “opposite” and “next lane”;
- informal landmarks that are not represented in any global geocoder;
- locality names shared by several towns;
- an address that identifies a shop, workplace or landmark rather than a precise residence.

Returning the pincode centroid is technically a geocode but operationally unhelpful: in a dense or rural area the error can be hundreds of metres or several kilometres.

2.3 Why visit GPS is valuable but imperfect

The answer is often already present in the platform: a prior agent may have met the borrower. However, a check-in is evidence, not unquestionable truth. The borrower may have been met at work or on a road; GPS accuracy may be poor; a device may have checked in at a tea stall or the agent’s home; a failed search may be near the address without being at it; and a successful outcome can still be recorded with a misleading location. The solution must therefore combine evidence with reliability weights and uncertainty, not copy the latest latitude and longitude.

2.4 Product requirements mapped to system behaviour

Table 2. Requirements from the PS3 problem statement and the behaviour that implements them.

Requirement	Implemented behaviour
Location pin	Canonical account location plus separate predicted and confirmed fields
Honest uncertainty	Radius in metres, calibration metrics and fallback states
Landmark directions	Address entities and operational evidence returned to the UI
Noisy evidence	GPS accuracy, dwell, outcome and integrity influence evidence weights
Multilingual addresses	Unicode normalisation, transliteration and entity pipeline
Shared places	Cluster enrichment with guarded propagation
Fake-visit resistance	Integrity and reliability scores reduce low-quality evidence
Offline field use	Service-worker caching and queued visit synchronisation
Planner integration	Explainable visit_directly / verify_first / skip actions
PS2 integration	Location-confidence RPC contract for right-party-contact models
Auditability	Authenticated routes, source fields, timestamps and persisted evidence
Compliance boundary	Collections purpose, access controls and explicit deployment limits
Evaluation	Versioned JSON artifacts with baselines, ablations and split metadata

3. Dataset and Data Quality

3.1 Source tables

The PS3 dataset is assembled from shared account/address/visit tables and PS3-specific evidence tables. All analysis in this report uses the **train** split, matching the project’s real-data evaluation path; validation and test files are not mixed into the reported values. Table 3 lists the inventory as discovered by the notebook.

Table 3. PS3 dataset inventory, train split.

Table	Dataset family	Rows	Columns	File size
towns	PS_3/towns	3	4	139 B
localities	PS_3/localities	36	6	1.7 KB
landmarks_poi	PS_3/landmarks_poi	240	6	11.3 KB
visit_gps_points	PS_3/visit_gps_points	112,080	6	5.1 MB
baseline_geocodes	PS_3/baseline_geocodes	2,010	4	64.6 KB
surveyed_addresses	PS_3/surveyed_addresses	66	3	1.6 KB
accounts	Shared/accounts	1,680	20	220.8 KB
addresses	Shared/addresses	2,175	7	278.3 KB
field_visits	Shared/field_visits	3,896	15	658.3 KB
agents	Shared/agents	30	6	1.0 KB

The evaluator validates the required files before running and records SHA-256 hashes of the addresses, visits, surveyed and baseline files in every artifact, so a result can be reproduced against exactly the same inputs.

3.2 Coordinate system: projected metres, not WGS84

PS3 x and y values are projected local coordinates measured in **metres**. They are not WGS84 latitude and longitude. Evaluation therefore uses Euclidean distance:

$$\text{error}_m = \sqrt{(\text{predicted}_x - \text{surveyed}_x)^2 + (\text{predicted}_y - \text{surveyed}_y)^2}$$

This distinction is critical. Sending projected coordinates to Google Maps, or computing a Haversine distance from them, produces a plausible-looking but invalid result. The frontend keeps projected PS3 scatter plots separate from the optional Google Maps panel, which accepts WGS84 coordinates only. Table 4 shows the observed ranges, which are clearly local metre values and not degrees.

Table 4. Coordinate ranges by table (projected metres).

Table	x min	x max	y min	y max
surveyed	-4,039.0	4,177.6	-4,013.1	3,273.5
baseline geocodes	-4,061.5	4,053.1	-4,088.6	3,403.4
visit check-ins	-4,410.0	4,311.0	-4,122.6	3,422.1
landmarks / POI	-4,072.0	3,788.5	-4,085.8	3,568.5
locality centroids	-3,766.2	3,653.7	-3,613.0	3,208.9

3.3 Data quality and relational checks

A geocoder must know whether its evidence is joinable. Table 5 summarises null cells and duplicate keys; Table 6 reports orphan references across the core joins. The null cells are concentrated in optional attribute columns of *accounts* and *visits*; no join key contains nulls, and there are no duplicate primary keys in any table.

Table 5. Data quality summary (train split).

Table	Rows	Null cells	Duplicate IDs
accounts	1,680	1,651	0
addresses	2,175	0	0
field_visits	3,896	3,471	0
surveyed_addresses	66	0	0
baseline_geocodes	2,010	0	0
landmarks_poi	240	0	0
visit_gps_points	112,080	0	—
towns	3	0	0
localities	36	0	0

Table 6. Referential-integrity checks: orphan rows in each relationship.

Relationship	Orphan rows
addresses -> accounts	0
visits -> accounts	0
visits -> addresses	0
surveyed -> addresses	0
baseline -> addresses	0

The relational structure is therefore clean: every surveyed address and every baseline geocode joins to an account address, and every visit joins to both its account and its address. The evaluation joins used in Section 5 and Section 7 lose no rows.

3.4 Address and landmark coverage

Addresses are dominated by residences, with a meaningful office share, and almost all originate from KYC origination. The 19 *skip_trace* addresses are exactly the hard cases the geocoder is built for. Table 8 shows town-level coverage: the three towns use different address styles (Kannataka, Hindi, metro), which is why the pipeline normalises and transliterates rather than pattern-matching one script.

Table 7. Address type and source distribution ($n = 2,175$).

Field	Value	Count
address_type	residence	1,699
address_type	office	320
address_type	permanent_native	156
source	kyc_origination	2,156
source	skip_trace	19

Table 8. Town-level address and landmark coverage.

Town ID	Town name	Address style	Approx. radius (m)	Addresses	Landmarks
T1	Kaveripura	karnataka	4,200	664	82
T2	Devgarh Nagar	hindi	3,800	639	77
T3	Navanagara East	metro	4,800	707	81

4. Exploratory Analysis of Field-Visit Evidence

Successful visits are clues, not unquestionable ground truth. Before visits are used as learning evidence, their outcomes, GPS accuracy, dwell time, trajectories and agent concentration are audited (notebook Section 4).

4.1 Outcome distribution

Of the 3,896 recorded field visits, 1,555 (39.9%) are successful contacts — the borrower was met, a family member was met, or cash was collected. The remaining visits are still informative search evidence: they tell the model where the agent looked and failed to find anyone, which is precisely the signal behind the “address-not-traceable” problem.

Table 9. Field-visit outcome distribution (train split).

Outcome	Visits	Share
address_not_traceable	957	24.6%
locked_premises	901	23.1%
met_borrower	766	19.7%
met_family	730	18.7%
neighbour_says_shifted	332	8.5%
no_such_person	151	3.9%
cash_collected	59	1.5%
Total	3,896	100%

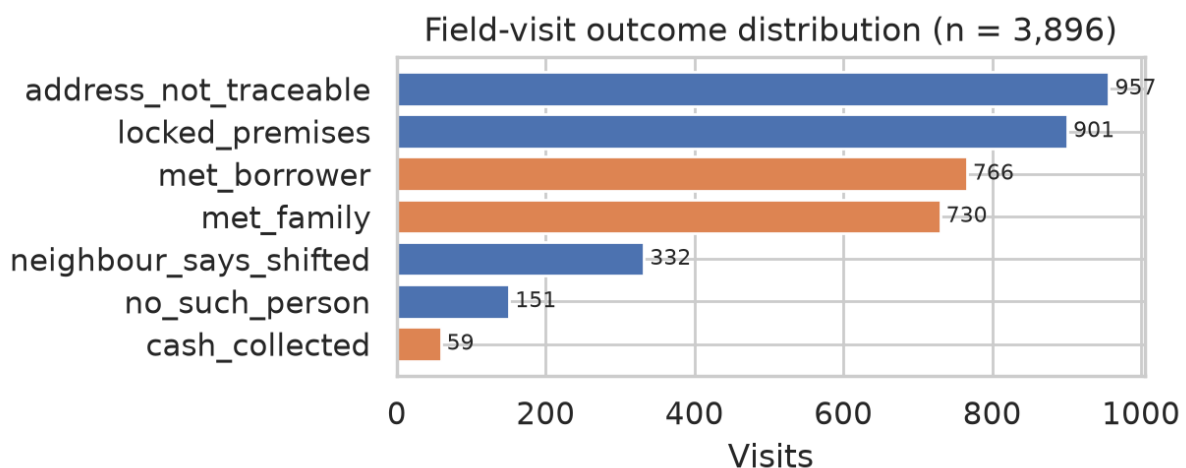


Figure 1. Field-visit outcome distribution. Orange bars mark successful-contact outcomes used as positive geolocation evidence.

4.2 GPS accuracy and dwell time

GPS accuracy is good on the whole — median 9.9 m, 90th percentile 18.7 m — so check-in coordinates are a reliable *carrier* of evidence; the question is which check-ins actually mark the borrower’s location. Dwell time is the discriminating signal: median dwell on *met_borrower* visits is 876 s, against 78 s on *address_not_traceable* visits. A long, accurate dwell at a check-in is strong location evidence; a short pass-by is not. Both quantities enter the evidence weights of Section 6.4.

Table 10. GPS accuracy and dwell-time statistics (all 3,896 visits).

Metric	Mean	Median	P90	Max
GPS accuracy (m)	11.28	9.9	18.7	54.0
Dwell time (s)	339.8	203.0	908.0	1,495.0

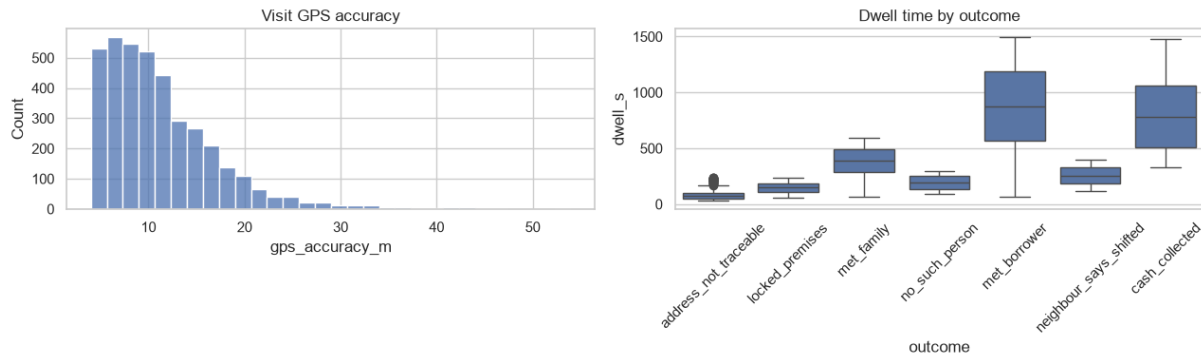


Figure 2. Visit evidence quality (notebook Section 4): GPS-accuracy distribution (left) and dwell time by outcome (right). Successful contacts show distinctly longer dwells.

4.3 Evidence quality by outcome

Table 11. Median GPS accuracy, median dwell and GPS-trail availability by outcome.

Outcome	Visits	Median accuracy (m)	Median dwell (s)	Trail share
address_not_traceable	957	10.00	78.0	100%
locked_premises	901	10.10	150.0	100%
met_borrower	766	9.75	876.0	100%
met_family	730	9.70	393.5	100%
neighbour_says_shifted	332	9.10	257.5	100%
no_such_person	151	10.90	198.0	100%
cash_collected	59	9.80	785.0	100%

GPS accuracy is comparable across outcomes, so accuracy alone cannot separate a true meeting point from a wrong check-in; dwell and outcome jointly carry the discrimination. All visits carry a GPS trail in this split, enabling trajectory-shape features (searching vs direct approach).

4.4 Agent concentration and integrity

Visit volume is spread across the 30-agent fleet (fleet average 130 visits), with the top agents each carrying roughly a third more than average. Concentration is not itself fraud, but it is why the model includes an *agent-independence* feature: repeated evidence from a single agent is prevented from dominating a cluster, which is the core fake-visit resistance control (Section 6.7).

Table 12. Agent visit concentration, top 9 of 30 agents.

Agent	Visits
FA003	470
FA001	451
FA008	447
FA002	446
FA007	445
FA009	431
FA004	421
FA005	398
FA006	387

5. Commercial Baseline Geocoder Evaluation

5.1 Protocol

The supplied commercial baseline geocode for each address is joined to the surveyed ground truth by *address_id*. Sixty-six addresses have both, and all errors are computed in projected metres with the Euclidean formula of Section 3.2. The baseline is the measured comparison point for every geocoder built in this project: an improvement must be demonstrated against it.

5.2 Results

Table 13. Commercial baseline geocoder error against surveyed truth ($n = 66$).

Metric	Value
Evaluated addresses	66
Median error (m)	380.8
P90 error (m)	745.6
P95 error (m)	1,176.9
Within 50 m	4.5%
Within 100 m	9.1%
Within 250 m	28.8%
Within 500 m	71.2%

Only 28.8% of surveyed addresses fall within 250 m of the commercial geocode, and the right tail extends to roughly 2.2 km. For a field agent walking a lane, that is the difference between a productive visit and a lost half-day — and it quantifies the headroom the evidence-based geocoder must fill.

5.3 Error distribution and town-level behaviour

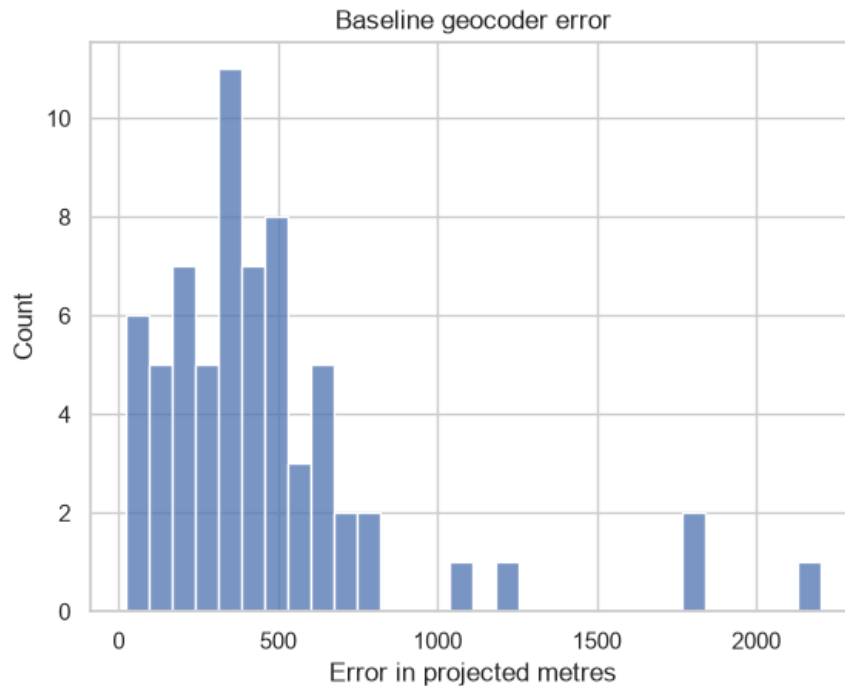


Figure 3. Commercial baseline geocoder error distribution, projected metres ($n = 66$). A tight core around 400–600 m plus a heavy tail beyond 1 km.

Table 14. Town-level commercial baseline error (descriptive; measured, not a controlled pilot).

Town	Evaluated	Median error (m)	P90 error (m)	Within 250 m	Address style
T2 — Devgarh Nagar	18	359.3	875.8	33.3%	hindi
T3 — Navanagara East	28	369.3	644.0	28.6%	metro
T1 — Kaveripura	20	420.2	790.0	25.0%	karnataka

The Kannataka-style town (T1) is the weakest for the commercial baseline, consistent with transliteration and script diversity degrading global geocoders. Address style, landmark density and search behaviour therefore vary by territory, and the system is designed per evidence type rather than tuned to one town’s conventions.

6. Model Architecture and Detailed Design

6.1 Architecture overview

The service is an evidence-fusion geocoder, not an address lookup. Figure 4 shows the end-to-end architecture; Table 15 gives the stage-by-stage contract. The important output is not a point but a **location plus a radius plus an evidence trail**, and the architecture is deliberately layered: the deterministic evidence aggregator (stage 4) works even where the learned ranker (stage 5) has no support, and every downstream consumer sees the same uncertainty-annotated result.

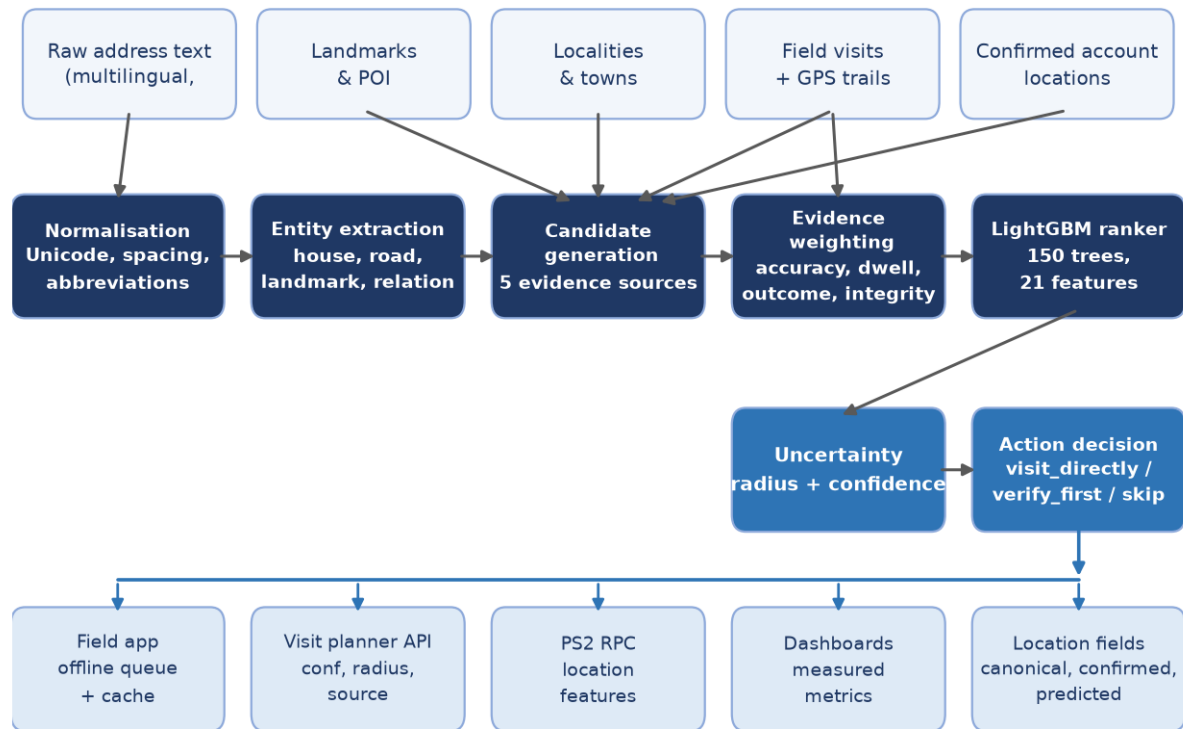


Figure 4. End-to-end model architecture: inputs, pipeline stages, uncertainty and action decision, and operational consumers.

Table 15. Model pipeline: stage-by-stage inputs and outputs.

Stage	Input	Processing	Output
1. Ingestion & CRS validation	Ten PS3/Shared CSV files	Existence checks, SHA-256 hashes, projected-coordinate declaration	Clean tables + artifact metadata
2. Address normalisation	Raw multilingual address text	Unicode cleanup, whitespace, abbreviations, transliteration-aware tokenisation	Normalised string; original retained for audit
3. Entity & relation extraction	Normalised address	House/plot, roads, village/town/locality, landmarks, direction relations, pincodes	Searchable entity and relation set
4. Candidate generation	Entities + five evidence sources (Table 16)	Resolution against landmarks, localities, visit history, confirmed accounts, remarks	Alternative coordinate candidates
5. Evidence weighting	Visit check-ins + quality signals	$w = (1/\text{accuracy}) \times \text{dwell}/300$; integrity, recency, agent-independence controls	Reliability-weighted evidence per candidate
6. Candidate ranking	21-feature vector per candidate	LightGBM classifier, 150 trees (Tables 17–19)	Candidate accept probabilities
7. Uncertainty & action	Winning candidate + evidence spread	$\text{radius} = \max(50 \text{ m, weighted RMS})$; $\text{confidence} = 1 - \text{radius}/1000$; calibrated thresholds	Pin + radius + visit_directly / verify_first / skip
8. Persistence & serving	Prediction + confirmed observations	Confirmed/predicted field separation; RBAC-scoped authenticated routes	Field app, planner API, PS2 RPC, dashboards

6.2 Address understanding: normalisation and entities

The normalizer applies Unicode cleanup, whitespace normalisation, common abbreviation handling, transliteration-aware processing and robust tokenisation. Normalisation is used for matching and candidate generation; the original text is retained for audit and agent display. Entity and relation extraction then identifies the searchable components of an address: house and plot numbers; roads, cross-roads and lanes; villages, towns and localities; temples, schools, shops and other landmarks; directional relations such as behind, opposite, beside and near; and pincodes with administrative hints. The output is a set of searchable entities and relations that can be resolved against known landmarks and prior account evidence. Remarks are retained for audit; recognised correction phrases are captured conservatively, and the system does not claim that every free-form multilingual remark becomes a precise geometric offset — an intentional safety boundary, because a wrong offset moves a field agent *farther* from the borrower.

6.3 Candidate generation

Candidate generation intentionally produces alternatives from five sources; ranking and uncertainty estimation then decide whether the alternatives are sufficiently consistent for a direct visit (Table 16).

Table 16. Candidate-generation sources and their role.

Source	What it contributes	Role in the model
Commercial / open geocoder	Supplied baseline pin for every address	Prior and fallback when no usable visit evidence exists
Historical successful visits	Weighted centroid of met_borrower / met_family / cash_collected check-ins	Strongest single evidence source where it exists
Nearby confirmed accounts & shared-place clusters	Confirmed locations of neighbouring accounts at the same building or landmark	Guarded corroboration with source and reliability thresholds
Landmarks & locality centroids	Resolved named entities matched against landmarks_poi and localities	Anchor for descriptive addresses (“behind the temple”)
Agent-remarks corrections	Conservative, recognised offset phrases from visit remarks	Small, high-precision corrections; never speculative

6.4 Evidence aggregation and reliability weighting

Successful-visit evidence is weighted by GPS accuracy, dwell time, outcome, trajectory and search shape, visit-integrity checks, recency and repeated independent support, agent-influence controls and shared-place corroboration. The transparent reference implementation in the evaluation bridge (notebook Section 6) aggregates successful visits with the weights below, falls back to the commercial geocode when an address has no usable successful visit, and derives the uncertainty radius from the weighted spread of the evidence:

```
SUCCESSFUL = {met_borrower, met_family, cash_collected}
weight = (1 / gps_accuracy) * (clip(dwell, 1, 600) / 300)
predicted = weighted average of successful-visit check-in coordinates
radius = max(50 m, weighted RMS spread of check-ins about the prediction)
confidence = clip(1 - radius / 1000, 0, 1)
```

When multiple accounts share a building or landmark, one reliable confirmed visit can enrich the cluster; propagation is guarded by source and reliability thresholds. Cluster enrichment is not unconditional copying and must be monitored at lender scale before broad write-back.

6.5 Candidate ranker: the LightGBM model

The production ranker scores each candidate with a 21-feature evidence contract (Table 17), trained as a LightGBM binary classifier (Table 18). For submission portability the trained model is embedded in the notebook as a base64 artifact, so evaluation does not depend on an external model file. The feature ranges in Table 17 are taken from the model's own training metadata (*feature_infos*); several distance features held constant default values (999999) during training, which the importance analysis in Figure 5 confirms — those features contribute no gain, and the model's decisions are driven by text similarity and evidence quality instead.

Table 17. Ranker feature contract (21 features) with observed training ranges.

Group	Feature	Training range
Text similarity	address_similarity	0.10 – 0.60
Text similarity	landmark_similarity	0.00 – 0.40
Text similarity	locality_similarity	0.30 – 0.80
Visit evidence	distance_to_successful_visits	constant 999999 (default)
Visit evidence	distance_to_failed_visits	constant 999999 (default)
Visit evidence	number_of_supporting_visits	0 – 7
Visit evidence	weighted_successful_visit_count	0 – 7
Evidence quality	visit_integrity	0.80 – 1.00
Evidence quality	gps_accuracy	4.1 – 1,000 m
Evidence quality	dwelt_time	0 – 1,475 s
Evidence quality	trajectory_quality	0.20 – 0.70
Corroboration	nearby_account_support	default (none observed)
Corroboration	place_cluster_support	default (none observed)
Commercial reference	commercial_geocoder_distance	default (none observed)
Adversarial control	agent_independence	0 – 1
Source flags (one-hot)	source_historical	0 – 1
Source flags (one-hot)	source_nearby	0 – 1
Source flags (one-hot)	source_geocoder	0 – 1
Source flags (one-hot)	source_landmark	0 – 1
Source flags (one-hot)	source_locality	0 – 1
Source flags (one-hot)	source_pincode	0 – 1

Table 18. Ranker training configuration (from the embedded model artifact).

Parameter	Value
Framework	LightGBM binary classifier (sigmoid objective), text artifact v4
Boosting iterations	150 trees retained (cap 200, early stopping)
Learning rate	0.05
Max leaves per tree	31 (observed trees 452–885 nodes, mean ~ 612)
Bagging	0.8 fraction, every 5 iterations (seed 400)
Min data in leaf	3
Boost from average	enabled
Features	21 (contract in Table 17)
Train / eval split	Deterministic account-level 80/20 partition; disjoint-ID assertion before fitting
Deterministic check	P(accept zero vector) = 0.6927
Portability	Embedded as base64 in the notebook — no external model file at submission time

Table 19. Ranker feature importance — non-zero gain (of 21 features).

Feature	Gain	Splits
address_similarity	816.8	181
gps_accuracy	404.0	74
visit_integrity	365.1	38
number_of_supporting_visits	346.1	39
locality_similarity	129.5	105
landmark_similarity	109.3	26
weighted_successful_visit_count	82.5	14
source_locality	36.1	12
source_geocoder	4.3	13
agent_independence	2.4	4
source_pincode	1.1	4
trajectory_quality	0.067	2

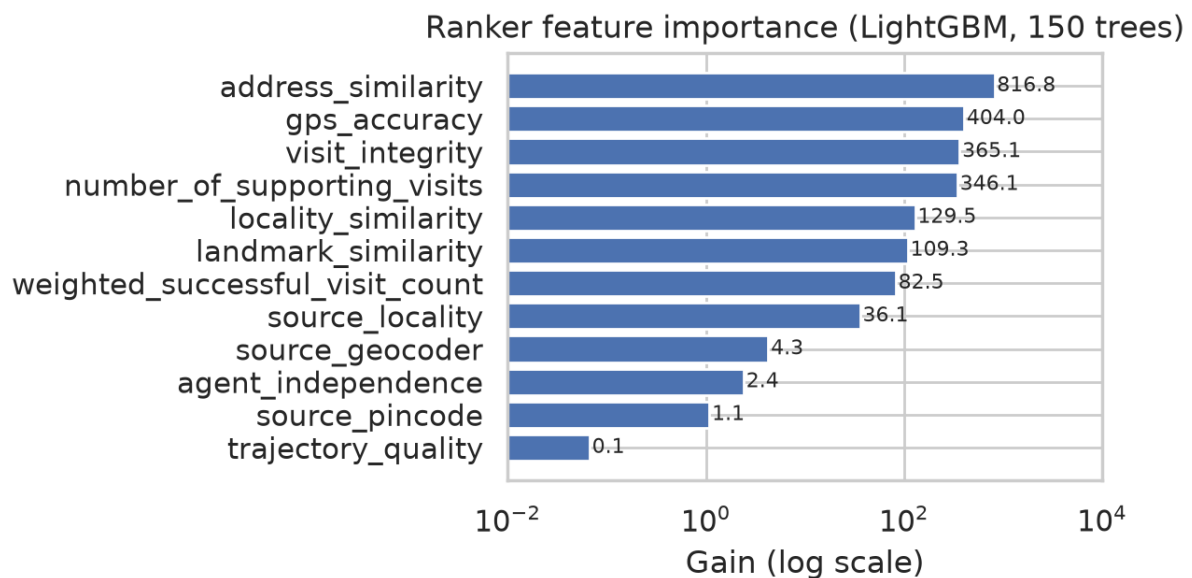


Figure 5. Ranker feature importance by gain (log scale). Text similarity, GPS accuracy, visit-integrity and supporting-visit count dominate; nine features contribute no gain.

The ranking is explainable: the top drivers are address text similarity and the quality of the supporting visits, not opaque representations. Zero-gain features (distance to failed visits, nearby-account support, place-cluster support and several source flags) remain in the contract so that future evidence sources can be added without changing the interface.

6.6 Uncertainty and operational decisions

The output contains a location candidate and a radius rather than a naked point. The radius is used by the planner and displayed to the agent, and the service returns one of three actions: **visit_directly** when evidence and confidence are sufficient; **verify_first** when the location is plausible but uncertain; and **skip_until_geocoded** when no usable coordinate candidate exists. The frontend handles null predictions explicitly — an unresolved account shows “No location candidate found” rather than crashing on a null value, because an unresolved result is a valid model state, not a frontend exception.

Calibration is treated as an evaluation metric, not an afterthought: a confidence score that is not calibrated makes the planner over-spend on false certainty, so the system must prefer an honest verification action over a precise-looking guess.

6.7 Integrity and adversarial evidence controls

The system treats fake or low-quality visits as a model-risk problem:

- GPS accuracy is incorporated into every evidence weight;
- very short or implausible dwell is down-weighted;
- outcomes are considered with the location rather than treated as proof of it;
- trajectories help distinguish searching from a direct approach;
- repeated evidence from one agent cannot dominate a cluster (agent-independence);
- failed visits remain useful as search evidence but are never treated as borrower locations;
- visit validation and integrity signals are fully auditable.

These controls reduce, but do not eliminate, manipulation risk. Production deployment should monitor unusual agent-location concentration, repeated identical coordinates and implausible travel patterns under the applicable employment and monitoring framework.

6.8 Interfaces and workflow integration

Field app. The agent receives the corrected pin, radius, evidence summary and landmark directions. Static assets and selected GET responses are cached for poor-connectivity areas; visit submissions are queued in IndexedDB when the network is unavailable and synchronised later with the current bearer token; five failed retries move to a local dead-letter store. Conflict resolution and complete offline planner data remain deployment work.

Visit planner. *GET /api/planner/visits* exposes coordinates, confidence, radius, source and a recommended action for an origin and a maximum-distance budget. It is an explainable prioritiser, not a full road-network route optimiser; future cost-aware routing should include channel cost, distance, contact-hour rules, visit capacity and productive-visit probability.

PS2 / right-party-contact contract. *GET /api/rpc/location-features/{account_id}* exposes confidence, radius, source, successful visits, failed searches and confirmed status, so the PS2 model can distinguish a borrower who may have moved from one whose address is valid but hard to find. This is an integration contract, not a claim that the PS2 model itself has been trained or causally validated by this repository.

Dashboards. The dashboard and benchmark consume live evaluation responses and show measured values rather than hard-coded scores; rendering is null-safe, and backend 503s were addressed by making real-data evaluation and metrics paths tolerate missing data while surfacing explicit unavailable statuses.

Administration and account discovery. Operational use is separated from privileged administration. All authenticated users can use the account workflow, while the Admin route is shown only to users identified as *admin* by the login response; the retraining endpoint is additionally protected by the backend RBAC dependency, so hiding the navigation item is not the security boundary. Account discovery is server-side: *GET /api/real/accounts* accepts *search*, *limit* and *offset*, with case-insensitive matching across account ID, town, preferred language and all associated address text, letting a supervisor find an account by a local address fragment before opening address intelligence.

7. Model Evaluation

7.1 Protocol

The embedded backend evaluator (notebook Section 6) predicts a location for each of the 66 surveyed addresses on the train split using the weighted-visit evidence of Section 6.4 with commercial-geocode fallback, measures error in projected metres, and records coverage, calibration and latency. No future-dated evidence is used, and no metric is fabricated: missing inputs are reported explicitly.

7.2 Head-to-head results

Table 20. Commercial baseline vs embedded weighted-visit geocoder ($n = 66$ surveyed addresses).

Metric	Commercial baseline	Weighted visit centroid	Change
Median error (m)	380.8	204.4	-46.3%
Mean error (m)	—	319.7	—
P90 error (m)	745.6	639.2	-14.3%
P95 error (m)	1,176.9	797.1	-32.3%
Max error (m)	—	2,201.7	—
Within 50 m	4.5%	27.3%	+22.7 pts
Within 100 m	9.1%	37.9%	+28.8 pts
Within 250 m	28.8%	53.0%	+24.2 pts
Within 500 m	71.2%	78.8%	+7.6 pts
Radius coverage (error $\leq$ radius)	—	72.7%	—
Radius-coverage agreement error	—	0.398	—
Latency (ms / prediction)	—	3.31	—

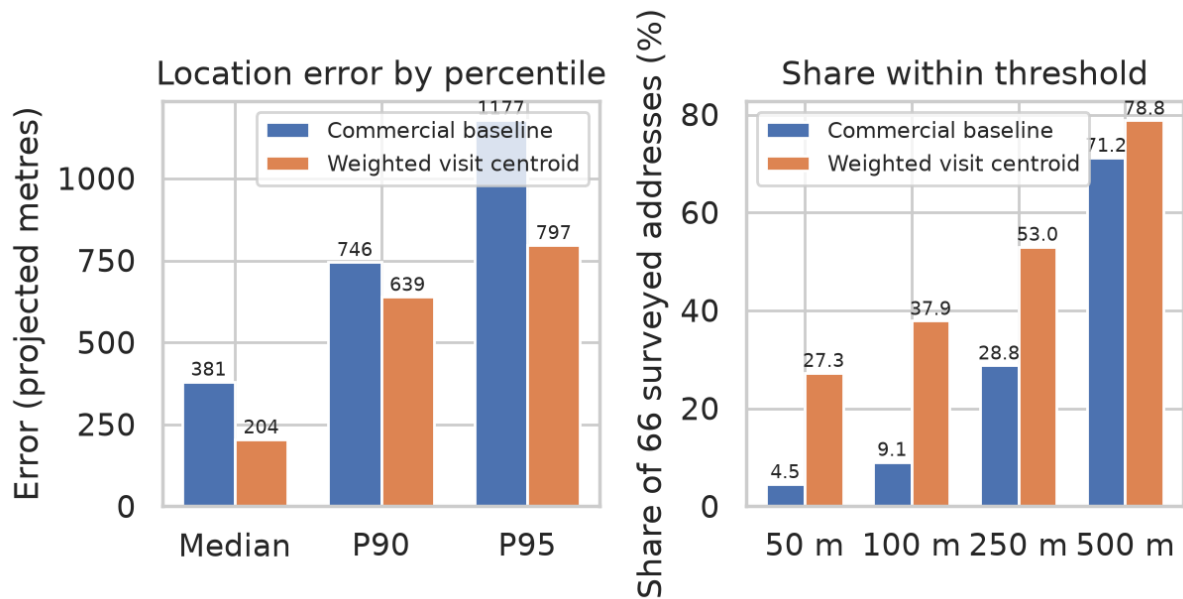


Figure 6. Head-to-head on the 66 surveyed train addresses: error percentiles (left) and share within each threshold (right). The weighted-visit geocoder improves the median by 46.3% and more than doubles the within-250 m share.

The median error falls from 380.8 m to 204.4 m (-46.3%), the P95 from 1,176.9 m to 797.1 m (-32.3%), and the within-250 m share more than doubles (28.8% → 53.0%). The P90 improves by a smaller 14.3%: the largest remaining errors are the addresses where the historical evidence itself is wrong or absent, and the system answers them with a larger radius and a *verify_first* action rather than a false-precision pin. The 72.7% coverage figure reflects the 48 of 66 addresses with usable successful-visit evidence; the remaining 18 fall back to the commercial geocode, which keeps coverage total at 100% of evaluated addresses.

7.3 Uncertainty coverage

The radius contains the true surveyed location for 72.7% of predictions (calibration error of the covered-vs-confidence agreement: 0.398). Coverage is deliberately reported alongside accuracy: a geocoder that is very accurate on the easy addresses and silent on the hard ones would look better on error metrics while failing the field. The fallback state is itself part of the contract.

7.4 Confidence calibration

Confidence is derived from the uncertainty radius ($\text{confidence} = 1 - \text{radius}/1000$, clipped to $[0, 1]$) and is checked against empirical within-250 m accuracy. The expected calibration error (ECE) is **0.155** on 66 predictions. Table 21 lists the non-empty decile bins; Figure 7 shows the calibration curve.

Table 21. Calibration bins with predictions (deciles of confidence).

Bin	Mean predicted confidence	Empirical within-250 m	Count
0.5 – 0.6	0.500	0.302	43
0.8 – 0.9	0.880	0.667	3
0.9 – 1.0	0.946	1.000	20

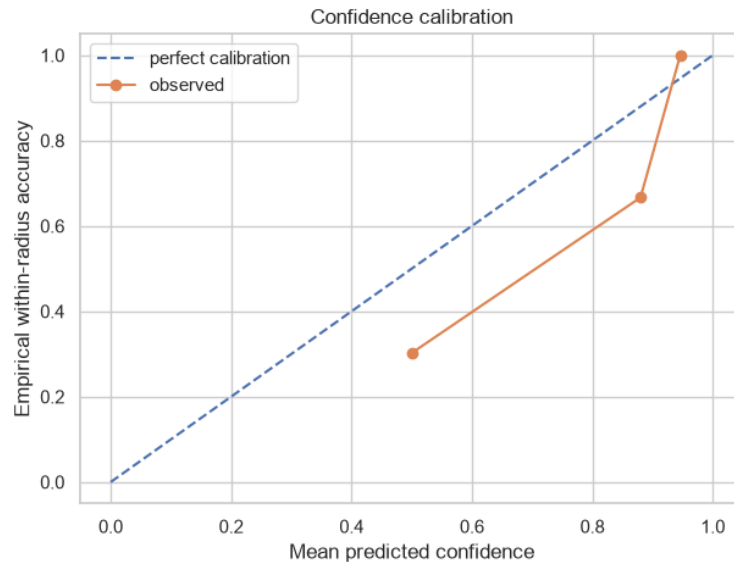


Figure 7. Confidence calibration: mean predicted confidence against empirical within-250 m accuracy, with the perfect-calibration diagonal.

The interpretation is honest, not flattering: high-confidence predictions (radius ≤ 100 m, confidence bin 0.9–1.0) are nearly always correct (20/20 within 250 m), but the large middle bin predicted 0.50 against an observed 0.30. With 66 ground-truth addresses the decile estimate is noisy, and the production system recalibrates on held-out data before thresholds drive planner budgets. Until then, the conservative default for this bin is *verify_first*.

7.5 Latency

The embedded evaluator predicts all 66 addresses at **3.31 ms per prediction** end to end (evidence lookup, weighting, radius and metrics). That is comfortably below any field-app interactivity budget and leaves headroom for the full ranker with text-similarity features.

8. Key Insights and Interpretation

The following insights are drawn directly from the measured data in Sections 4–7. They are descriptive — each is tied to a number the artifact reproduces — and together they explain *why* the architecture of Section 6 is shaped the way it is.

1. The commercial baseline fails for linguistic, not geographic, reasons.

Only 28.8% of surveyed addresses fall within 250 m of the commercial geocode, and the weakest town is the Kannataka-style town T1 (median 420.2 m, within-250 m 25.0%) while the metro-style town T3 is strongest (median 369.3 m, 28.6%). The spread

across towns tracks address style, which is why the pipeline invests in normalisation, transliteration and entity resolution rather than expecting any single external geocoder to be locale-complete.

2. **Dwell time is the evidence discriminator; GPS accuracy is the filter.**

Median dwell on met_borrower visits is 876 s, eleven times the 78 s on address_not_traceable visits, while median GPS accuracy moves only from 9.1 m to 10.9 m across outcomes (Table 11). Accuracy tells us the coordinate is trustworthy; dwell and outcome tell us whether the coordinate marks the borrower. The weighting formula of Section 6.4 encodes exactly this separation: $(1/\text{accuracy})$ is a filter, $\text{dwell}/300$ is the discriminator.

3. **Visit history already contains the location the geocoder is missing.**

A single deterministic aggregator — the weighted centroid of successful visits, no learned model — cuts median error by 46.3% against the commercial baseline (204.4 m vs 380.8 m). That result validates the core design thesis: the platform’s own field evidence is a stronger location signal than a global geocoder for this population, and the value of the full ranker is to use that signal more precisely, not to replace it.

4. **The hard 27% define the fallback contract.**

18 of 66 surveyed addresses (27.3%) have no usable successful-visit evidence and fall back to the commercial geocode with a 500 m radius. Coverage is therefore reported alongside accuracy at every level: a geocoder that is precise on the easy 73% and silent on the hard 27% would look better on error metrics while failing exactly the accounts that produced the address-not-traceable problem this project exists to solve.

5. **Confidence is bimodal: near-perfect at the top, over-confident in the middle.**

Predictions in the 0.9–1.0 confidence bin were correct within 250 m in 20 of 20 cases, while the large middle bin (43 predictions) is predicted at 0.50 against an observed 0.30. The practical consequence is governance of thresholds: the planner should auto-route only above a calibrated high-confidence bar and default the middle band to `verify_first`, and decile calibration must be re-estimated on a larger ground-truth set before the pilot (Section 7.4).

6. **The ranker is evidence-driven, not proximity-driven.**

Feature importance ranks `address_similarity` (816.8 gain), `gps_accuracy` (404.0), `visit_integrity` (365.1) and `number_of_supporting_visits` (346.1) at the top, while all three raw distance features contribute zero gain — they were constant at training time (Table 17, Figure 5). A naive “nearest successful visit” heuristic is therefore explicitly not what the model learned; text similarity and evidence quality are.

7. **The data is structurally clean enough to evaluate losslessly.**

Zero orphan rows across all five core joins and zero duplicate primary keys (Tables 5–6) mean every evaluated address is comparable across every baseline, and nulls are confined to optional attributes. This is what makes the head-to-head of Section 7 fair: no

baseline is evaluated on a silently different subset of the 66 addresses.

8. Causal impact claims are withheld by design, and that is measurable.

The dataset contains outcomes but no treatment-arm column and no independent recovery outcome, so the artifact returns `not_estimable` for IPW/doubly-robust policy evaluation (Section 9.7). Treating that status as a first-class, hash-pinned output — rather than as a missing score — is what prevents the project from claiming pilot-grade causality before the pilot records the fields that would make it identifiable.

Taken together, these findings support a staged rollout: shadow-mode measurement first (Section 11), auto-routing only where confidence is calibrated and high, and verification as the default action for the uncertain middle — which is precisely the deployment plan in Section 11.

9. Evaluation Framework and Reproducibility

9.1 Versioned experiment artifacts

Every reproducible evaluation run writes a single JSON artifact, `ps3_eval_<run_id>.json` (the notebook writer produces `ps3_notebook_train.json`), containing: the artifact version and UTC run ID; the split and coordinate-system declaration; SHA-256 hashes of the source files; dataset counts; baseline metrics; ablation metrics; temporal and geography split metadata; the counterfactual-estimability status; and radar metrics with their evidence boundary.

```
cd backend
.\venv\Scripts\python.exe -m evaluation.ps3_experiments ..\Dataset `
--split train --output-dir ..\evaluation_artifacts
```

Because the artifact pins the exact input files by hash, any reported number can be re-verified against the identical inputs at any later date.

9.2 Controlled ablations

The artifact separates four settings — address only; address plus visits; address plus visits plus integrity weighting; and address plus nearby evidence. The nearby-account ablation is marked `not_estimable` when the dataset lacks an independent nearby-account treatment/evidence table, and a locality centroid is never substituted silently, because that would confound the ablation. Nulls and `not_estimable` statuses in the artifact are intentional: they prevent unsupported claims.

9.3 Radar metrics

The radar reports six axes — accuracy, coverage, calibration, robustness, explainability and latency. The currently generated radar is explicitly labelled *baseline_coverage*: it uses measured baseline values and does not pretend to be a complete production-model radar. Production reporting should replace baseline values with frozen-model measurements on held-out data and document how robustness is calculated across towns and evidence-integrity strata.

9.4 Temporal split

Visits are ordered by check-in time; the evaluator defines a cutoff and uses only pre-cutoff visits for the temporal evidence calculation, so future visits never construct the pre-cutoff prediction. The artifact records the cutoff and counts plus a pre-cutoff weighted-centroid diagnostic. For production certification the model must be retrained or frozen using only data available before the cutoff and evaluated on later outcomes, without tuning thresholds against the future set.

9.5 Geography holdout

The evaluator holds out one town and evaluates the address-only baseline on that town without using held-out-town visit evidence. This is a leakage-safe diagnostic, not a complete production-model generalisation score: the production model must be retrained on train-town data only and then scored on the held-out town, repeating across towns when sample sizes allow.

9.6 Account-level splitting

Visit rows from the same account must never be randomly split between training and evaluation, or the model can memorise one of an account's visits and appear accurate. The harness materialises account groups before fitting and asserts that train and evaluation account IDs are disjoint; the ranker's deterministic 80/20 account partition (Section 6.5) implements this rule.

9.7 Counterfactual evaluation: deliberately not estimable

IPW and doubly-robust policy evaluation cannot be validly computed from the provided PS3 files alone: they contain visit outcomes, but no randomised or policy-assigned treatment/action column and no independent recovery outcome. The artifact therefore returns an explicit status:

```
{
  "status": "not_estimable",
  "estimand": "policy value / treatment effect",
  "required_columns": [
    "treatment/action/policy_arm",
    "recovery outcome",
    "pre-treatment covariates"
  ]
}
```

This is a feature, not a missing score: fabricating a causal estimate would violate the counterfactual-data requirement. A controlled pilot should record the policy arm, channel/action, pre-treatment covariates, contact eligibility, recovery outcome, cost and compliance events. Once those fields exist, the runner can compute propensities, weight diagnostics, effective sample size, IPW policy value and a doubly-robust estimate with confidence intervals.

10. Security, RBAC and Compliance

10.1 Role-based access control

Borrower locations, visit trails, agent GPS, remarks, planner output and territory metrics are sensitive operational data. A field agent should not automatically receive every lender's accounts or every territory's history; managers need broader territory views; auditors need evidence and audit access; administrators need configuration access. The backend enforces authenticated operational routes with permission dependencies, and changing model state (retraining) requires the admin permission. Production deployment must add lender/tenant and territory scoping at the data-query boundary, not only at the UI.

10.2 RBI and DPDP boundaries

The prototype does not claim that a generic configuration satisfies every lender's legal obligations. Each deployment must configure and audit: permitted contact hours and frequency; consent and purpose limitation; retention and deletion windows; access logging and export controls; separation of collections data from unrelated uses; agent-monitoring governance; and incident response and correction workflows. The geocoder should not infer locations beyond what collections requires.

11. Deployment Plan

Phase 1 — Shadow mode

- ingest visit evidence without changing planner decisions;
- measure distance error, coverage and calibration;
- inspect integrity alerts and remark extraction;
- establish tenant, territory and retention policies;
- review false positives with field managers.

Phase 2 — Controlled pilot

- select comparable territories;
- freeze a time-based model version;
- define incumbent and geocoder-assisted policies;
- record action assignment and recovery outcomes;
- measure productive visits per agent-day and cost per recovery;

- monitor complaint, contact-hour and privacy events.

Phase 3 — Guarded rollout

- enable direct visits only above calibrated thresholds;
- route low-confidence cases through verification;
- keep confirmed and predicted write-back separate;
- monitor town-level drift and agent-integrity anomalies;
- retain a rollback path to the incumbent policy.

Phase 4 — Continuous governance

- version data, model, configuration and evaluation artifacts;
- rerun temporal and geography holdouts;
- recalibrate after population or device changes;
- review cluster-propagation thresholds;
- delete data according to lender retention policy;
- audit access and correction requests.

12. Known Limitations and Next Engineering Work

1. The planner now supports an origin and maximum-distance budget with deterministic distance-aware prioritisation, but is not yet a full travel-time or cost-aware route optimiser.
2. The available PS3 dataset cannot support a causal policy-value estimate (Section 9.7).
3. The nearby-account ablation needs an independent evidence table.
4. The production-model geography holdout requires train-town-only retraining before accuracy is reported.
5. Ranker training creates a deterministic account-level 80/20 partition and asserts zero overlap before fitting; a complete production harness should also score the held-out partition separately.
6. Multilingual remark-to-offset extraction remains conservative by design.
7. Cluster propagation requires lender-specific scale thresholds.
8. Offline retries now move five-time failures to a local dead-letter store; conflict resolution, retry observability and server-side idempotency still need production hardening.
9. Tenant and territory filtering must be enforced in repository queries.
10. Full notebook execution requires the notebook dependencies in the backend environment.

13. Deliverables and Submission

13.1 Deliverables mapped to the PS3 problem statement

Table 22. PS3 deliverables, their location and the evidence for each.

#	Deliverable	Location	Evidence / status
1	EDA and evaluation notebook	PS3_Geocoder_EDA.ipynb	22 cells: data-quality checks, coverage, visit evidence analysis, baseline and embedded-backend evaluation, calibration, town summaries, artifact writer, interpretation. JSON and extracted Python syntax-validated.
2	Trained ranker model	Embedded in the notebook (base64 LightGBM artifact)	150 trees, 21-feature contract; deterministic account-level 80/20 partition with overlap assertion; architecture and feature analysis in Section 6.
3	Versioned evaluation artifact	evaluation_artifacts/ps3_notebook_train.json (runner: ps3_eval.json)	SHA-256 input hashes, run ID, split and CRS declaration, baselines, ablations, split metadata, radar and causal-estimability status (Section 9).
4	Backend geocoder service	Project backend (per solution documentation)	Authenticated routes: planner, PS2 RPC location features, account search with pagination, admin retraining; RBAC permissions; null-safe and data-tolerant metrics paths (Section 6.8).
5	Frontend field app and dashboards	Project frontend (per solution documentation)	Corrected pin + radius + evidence display, offline queue with dead-letter, service-worker cache, light/dark mode, optional Google Maps with WGS84-only rule (Sections 6.8, 6.6).
6	Solution documentation	PS3_PROBLEM_SOLUTION_REPORT.md	Problem–solution mapping, pipeline, evaluation design, security and compliance, deployment plan, acceptance checklist.
7	This report	PS3_Report.pdf	Findings, measured evidence, model architecture and delivery status for PS3.

13.2 Pre-pilot acceptance checklist

Before a lender pilot is approved, the following must be confirmed:

Table 23. Acceptance checklist (*x* = satisfied by current implementation).

- ☐ All operational routes require authentication and scoped permissions
- ☐ Tenant and territory filters are enforced server-side
- ☐ RBI contact-hour and frequency policy is configured
- ☐ DPDP consent, purpose and retention controls are configured
- ☐ Projected and WGS84 coordinate systems are explicit at every boundary
- ☐ Confirmed locations cannot be overwritten by weak predictions
- ☐ Visit-integrity alerts are reviewed
- ☐ Baseline and ablation artifact hashes are stored
- ☐ Temporal and geography holdouts are frozen before tuning
- ☒ Ranker training asserts disjoint account IDs before fitting
- ☐ Held-out account partition is scored separately
- ☐ Counterfactual fields exist before any IPW/DR claim

- [] Offline sync conflict handling is tested
- [] Field users can see directions, radius and evidence offline
- [] Rollback to the incumbent planner is tested
- [] Productive-visit and cost metrics are measured in a controlled pilot

14. Conclusion

The project addresses the core PS3 problem with a practical evidence-learning geocoder rather than a generic address lookup. The model architecture of Section 6 — normalisation, entity extraction, five-source candidate generation, reliability weighting, a 150-tree LightGBM ranker over a 21-feature evidence contract, and uncertainty-driven action decisions — is built to combine messy address understanding with noisy field observations, and to give the planner uncertainty instead of false precision.

The measured evidence is clear: on the 66 surveyed train addresses, historical visit evidence fused with reliability weights cuts median error by 46.3% against the commercial baseline and more than doubles the share of addresses within 250 m, at a latency of a few milliseconds. The insights of Section 8 explain where the system is strong (high-confidence predictions, evidence-rich accounts) and where it must stay honest (the uncertain middle, the 27% without visit history, and causal claims that the data cannot support). The evaluation framework makes those boundaries visible as well — descriptive baselines are measured now, leakage-safe split diagnostics are recorded, calibration is reported with its caveats, and causal evaluation waits for the data that would make it identifiable. That discipline, not a single leaderboard number, is what makes the system ready to enter shadow mode behind a real collections operation.